

KOFORIDUA TECHNICAL UNIVERSITY

FACULTY OF APPLIED SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE



HND COMPUTER SCIENCE WEEKEND

TOPIC

ELECTRONIC MEMO MANAGEMENT SYSTEM

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1.0. Background of the Project (Introduction)

Effective communication plays a critical role in ensuring smooth administrative operations and institutional effectiveness in higher educational institutions. Universities depend heavily on internal communication systems to transmit policies, academic instructions, administrative directives, and organizational updates across faculties, departments, and management units. At Koforidua Technical University (KTU), internal communication is largely facilitated through the use of official memoranda (memos), which serve as formal communication tools for disseminating important information within the institution.

Traditionally, many higher educational institutions rely on paper-based memo systems to manage internal communication processes. These systems involve manual preparation, printing, physical distribution, and storage of memos across departments. While this method has been used for decades, it presents numerous administrative and operational challenges. According to Ramoni (2019), manual memo handling systems often result in communication delays because documents must physically move from one office to another before reaching the intended recipients. Such delays can affect decision-making timelines and reduce institutional efficiency.

In addition to delays, paper-based memo systems increase the risk of document misplacement and loss. When memos are manually handled and stored, maintaining proper record-keeping becomes difficult, especially in institutions that generate large volumes of administrative documents. Research conducted by Afolabi and Abioye (2020) indicates that poor document management practices in organizations can lead to information retrieval difficulties, loss of institutional memory, and reduced accountability. In academic institutions, where documentation is essential for policy implementation and administrative transparency, such inefficiencies can negatively affect overall performance.

Furthermore, manual memo distribution lacks effective tracking and monitoring mechanisms. It is often difficult for administrators to confirm whether a memo has been delivered, received, or acknowledged by the intended recipient. This lack of transparency may create communication gaps, misunderstandings, and delays in executing institutional directives. Studies on electronic communication systems highlight that automated tracking and read-receipt functionalities significantly improve communication accountability and workflow monitoring (Olatokun & Adebayo, 2012).

The rapid advancement of Information and Communication Technology (ICT) has transformed organizational communication and document management practices globally. Many institutions are adopting electronic document management systems to improve efficiency, reduce paperwork, and enhance secure information storage. Electronic memo systems provide centralized platforms where users can create, distribute, archive, and retrieve memos efficiently. According to Dahlberg, Guo, and Ondrus (2015), digital communication platforms enhance operational effectiveness by enabling real-time information sharing, automated notifications, and improved data accessibility.

The adoption of electronic memo management systems has also contributed to improved administrative transparency and accountability in organizations. Digital systems allow authorized users to track memo circulation, verify delivery status, and maintain secure records of communication history. These systems support better collaboration among departments and improve institutional decision-making processes. Additionally, electronic memo systems

promote environmental sustainability by reducing paper consumption and supporting green institutional initiatives (Akanbi & Olalekan, 2021).

Despite the benefits associated with digital communication systems, many developing educational institutions still rely on manual memo processes due to infrastructure limitations, resistance to technological change, or lack of customized digital solutions. As administrative workloads continue to increase and institutional operations become more complex, the need for automated communication systems becomes more critical. Implementing an electronic Memo Management System at KTU will address existing communication challenges by providing a reliable and efficient platform for managing internal memos.

This project therefore proposes the design and development of a web-based Memo Management System that will automate memo creation, distribution, tracking, notification, and archival processes within KTU. The proposed system will integrate features such as targeted memo distribution, read-receipt tracking, email and SMS notifications, and secure digital storage. By transitioning from manual memo workflows to an electronic system, KTU will enhance communication efficiency, improve record management, and strengthen administrative accountability. The successful implementation of this system is expected to support the university's digital transformation agenda and improve the overall effectiveness of its internal communication processes.

2.0. Problem Statement

Internal communication plays a vital role in ensuring effective administrative operations in higher educational institutions. At Koforidua Technical University (KTU), internal communication is largely conducted through traditional paper-based memoranda, which present several operational challenges. One major issue is the delayed delivery of important notices and approvals due to manual preparation and physical distribution of memos. These delays can slow down decision-making processes and reduce administrative efficiency.

Another significant challenge is the frequent misplacement or loss of memos during circulation or storage. This disrupts workflow continuity and may result in departments failing to receive critical information. Additionally, the current memo system lacks effective tracking and auditing mechanisms, making it difficult to confirm whether memos have been delivered, received, or read by intended recipients. This reduces accountability and makes follow-ups challenging.

Furthermore, manual memo handling contributes to inefficient record-keeping practices, which complicate document retrieval and limit access to historical institutional information. As the volume of administrative documents increases, managing and retrieving archived memos becomes time-consuming and labor-intensive. These challenges collectively create communication bottlenecks, increase administrative workload, and reduce operational effectiveness, highlighting the need for an automated Memo Management System to improve communication efficiency and document management at KTU.

3.0. Aim of the Project

The aim of this project is to design and implement an electronic Memo Management System for Koforidua Technical University to improve internal communication, enhance document tracking, and ensure efficient memo storage and retrieval.

4.0. Objectives of the Study

4.1. Main Objective

The main objective of this study is to design and develop a Memo Management System that automates memo creation, distribution, tracking, and archiving processes within the university.

4.2. Specific Objectives

1. To design and develop a module that ensures memos are distributed to targeted users, departments, or organizational units.
2. To design and develop a module that implements tracking and read-receipt functionality to monitor memo circulation.
3. To design and develop a module that notifies users about memo updates and tracking status using SMS and email notifications.

5.0. Scope of the Project

This project focuses on the development of an electronic Memo Management System to support internal communication within the administrative units and departments of the university. The system will cover memo creation, approval, distribution, receipt confirmation, tracking, storage, categorization, and retrieval of memos. It will allow authorized staff to send memos electronically and monitor memo status in real time.

The project will be limited to internal institutional communication and will not initially include external stakeholder communication such as vendors, external organizations, or public announcements. Additionally, the system will focus on web-based accessibility for authorized university staff and administrators, with potential future expansion to mobile platforms and external communication integration.

6.0. Methodology (Tools & Technologies, Justification)

This project will adopt the Agile Software Development Methodology to guide the design and implementation of the Memo Management System for Koforidua Technical University. Agile methodology is selected because it promotes flexibility, continuous improvement, user involvement, and iterative system development. The methodology allows system features to be developed in phases, tested continuously, and improved based on feedback from users and stakeholders.

6.1 Agile Methodology Approach

The Agile methodology will be implemented through the following phases:

6.1.1. Requirement Gathering and Planning

This phase will involve collecting system requirements from university administrators, department heads, and registry staff. Interviews, observation, and document analysis will be used to understand current memo handling procedures, communication challenges, and user expectations. The gathered requirements will be analyzed and prioritized based on institutional needs. Agile planning will also involve breaking the project into smaller development cycles known as iterations or sprints.

6.1.2. System Design

During this phase, the overall structure of the Memo Management System will be designed. System models such as flowcharts, use case diagrams, database schema, and user interface prototypes will be developed. The system architecture will define how users interact with the system, how data flows between modules, and how security and authentication will be implemented. The design phase ensures that the system is scalable, user-friendly, and efficient.

6.1.3. Development and Implementation

The system will be developed incrementally based on prioritized requirements. Each sprint will focus on building specific modules such as memo creation, distribution, tracking, notification, and archiving. Continuous integration will be applied to ensure that developed modules function properly together. This iterative development approach allows for early detection of system issues and enables improvements throughout the development process.

6.1.4. Testing and Evaluation

Testing will be conducted continuously during development to ensure system reliability, security, and usability. Unit testing will verify individual system components, while integration testing will ensure modules work together effectively. User Acceptance Testing (UAT) will be conducted with selected university staff to evaluate system performance and user satisfaction. Feedback obtained during testing will be used to refine and improve the system.

6.1.5. Deployment and Maintenance

After successful testing, the system will be deployed for pilot use within selected departments. Training sessions will be conducted to familiarize users with system functionalities. Continuous maintenance and system updates will be performed to fix errors, improve performance, and introduce new features based on user feedback.

6.2 Tools and Technologies

The development of the Memo Management System will utilize modern web development technologies to ensure system reliability, scalability, and accessibility.

6.2.1. Database – MySQL

MySQL will be used as the relational database management system for storing memo records, user data, tracking information, and notification logs. MySQL is selected because it is reliable, secure, widely supported, and capable of handling large volumes of structured data efficiently.

6.2.2. Backend Development – PHP

PHP will be used as the server-side programming language for implementing system logic, authentication processes, memo processing, and database interaction. PHP is chosen because it integrates easily with MySQL, supports web-based applications, and provides strong community and documentation support.

6.2.3. Frontend Development – HTML, CSS, and JavaScript

HTML will be used to design the structure of the web pages, CSS will enhance the visual appearance and layout of the system interface, and JavaScript will provide interactive functionalities such as real-time notifications, form validation, and dynamic content updates. These technologies ensure the system is user-friendly, responsive, and accessible across different devices and browsers.

6.2.4. Server Environment – XAMPP Apache Server

The XAMPP Apache server will be used as the local development and testing environment. It provides an integrated platform that supports PHP execution, MySQL database management, and server configuration. XAMPP is selected because it simplifies system deployment, supports rapid testing, and provides a stable development environment.

6.3. Justification for the Selected Methodology and Tools

The Agile methodology is suitable for this project because it allows continuous user involvement and supports gradual system improvement through iterative development. Since the Memo Management System is intended to address real administrative challenges, Agile ensures that user feedback is incorporated throughout the development lifecycle, thereby improving system usability and effectiveness.

The selected technologies are cost-effective, widely used in web application development, and suitable for institutional systems. MySQL ensures efficient data storage and retrieval, PHP provides strong backend processing capabilities, and HTML, CSS, and JavaScript support the creation of an interactive and user-friendly interface. The use of XAMPP Apache server provides a reliable development and testing environment that supports rapid system implementation.

7.0. Limitation of the Study

This project focuses on developing a Memo Management System for internal communication within Koforidua Technical University. However, several limitations are anticipated. The system will initially be limited to internal administrative communication and will not support communication with external stakeholders such as vendors, government agencies, or the general public. Additionally, the system will be developed as a web-based platform and may require stable internet connectivity for effective operation, which could affect accessibility during network interruptions.

The project is also constrained by time and resource availability, which may limit the implementation of advanced features such as mobile application integration, advanced analytics, and full automation of approval workflows. Furthermore, user resistance to adopting new digital systems may affect early system acceptance and usage. Despite these limitations, the system will provide a strong foundation for future improvements and expansion.

8.0. Academic and Practical Relevance of the Project

Academically, this project contributes to research in information systems, software engineering, and digital document management. It provides practical experience in applying software development methodologies, database design, and web application development techniques to solve real-world institutional challenges. The project also supports academic research related to electronic communication systems and digital transformation in educational institutions.

Practically, the project aims to improve administrative efficiency by automating memo creation, distribution, and tracking processes. The system will enhance communication transparency, reduce paperwork, improve document storage and retrieval, and strengthen institutional accountability. The implementation of the system will also support digital transformation initiatives and promote the adoption of modern information management technologies within the university environment.

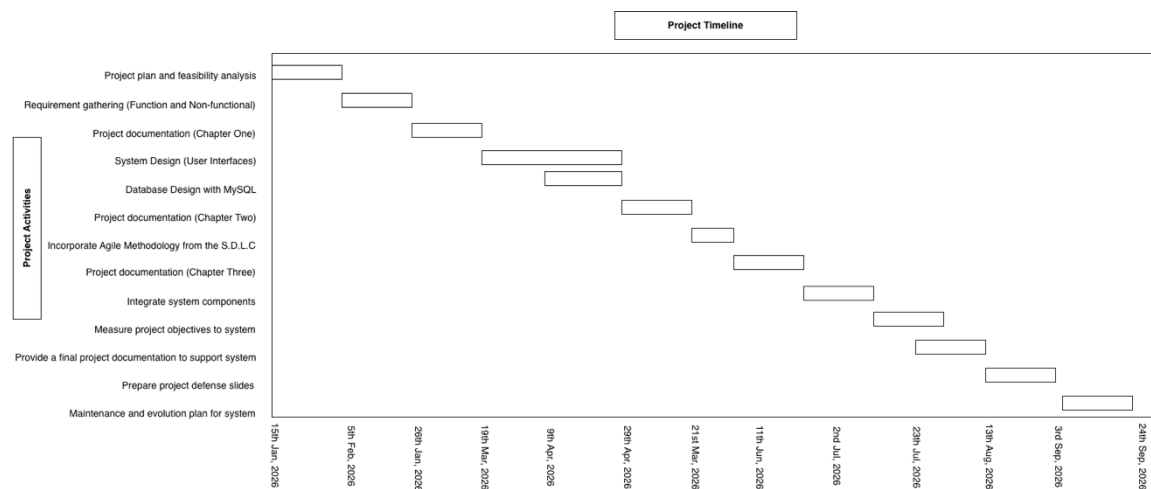
9. Beneficiaries of the Project

The Memo Management System will benefit several stakeholders within the university, including:

1. **University Management** – Will benefit from improved communication monitoring, faster decision-making, and enhanced administrative coordination.
2. **Administrative Staff** – Will experience reduced workload through automated memo handling, improved document tracking, and easier record retrieval.
3. **Academic Departments** – Will receive timely communication regarding academic activities, policies, and institutional updates.
4. **Registry and Records Offices** – Will benefit from improved document storage, organization, and retrieval processes.
5. **Students (Indirect Beneficiaries)** – Will benefit from improved administrative efficiency, leading to better service delivery and faster communication of institutional decisions.

10. Project Activity Planning

The project activity planning will be structured using a project timeline approach to ensure proper scheduling and monitoring of project tasks.



11. References (IEEE Referencing Style)

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